

Coupling Regimes, Resonance, and Synchronisation

Communications-Engineering Classification as a Structural Principle and its FFGFT
Connection

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Abstract

Communications engineering has a quantitative three-way classification of coupling regimes (undercritical, critical, overcritical) relative to the damping of the coupled subsystems. This document shows that this classification is an overlooked structural principle: the same mathematical structure recurs in quantum optics, particle physics, and synchronisation theory. The connection to FFGFT is established through ten verification scripts with explicit symbolic and numerical results.

Main results:

- [B] Dictionary two-circuit $\leftrightarrow$ 2 $\times$ 2 mixing (Hermitian + non-Hermitian, exceptional point $g_{EP} = |\gamma_1 - \gamma_2|/4$)
- [B] Exact duality $D^2 + V^2 = 1$ for all pure pointers; measurement transition as one-parameter continuum $\kappa = d/(2\sigma)$
- [K] $\kappa_{\text{mix}}(\theta_W)$ derived from ξ by composition (Doc. 323); screening negative (no simple constant)
- [B] GST texture $H_{12} = \sqrt{m_1 m_2}$ proven symbolically (Doc. 041 formulas at leading order)
- [K] $K_{\text{eff}} = 2\pi\xi \approx 8.4 \times 10^{-4}$ from Doc. 007/008; φ quantitatively outside all Arnold tongues

- Four check-points identified in Doc. 041 (author decision required)

.1 Motivation and Thesis

Theoretical physics treats couplings primarily via the magnitude of coupling constants. Communications engineering, by contrast, has an older and structurally richer classification: couplings are judged not absolutely but *relative to the damping of the coupled subsystems*. This yields a three-way split — undercritical, critical, overcritical — distinguishing not merely quantitative shifts but **qualitative changes of spectral structure**.

Thesis

This regime classification is an overlooked ordering principle. It recurs in quantum optics (strong coupling), particle physics (mixing, avoided crossing), and synchronisation theory (Kuramoto threshold, Arnold tongues) under different names. For FFGFT, in which particles are resonance states, it provides the natural language for when one state becomes two — and which frequency ratios lock or remain stably incommensurate.

.2 The Two-Circuit Formalism [E]

Two resonators with eigenfrequencies ω_1, ω_2 , quality factors Q_1, Q_2 and coupling coefficient k . Reference quantity: **critical coupling**

$$k_{\text{crit}} = \frac{1}{\sqrt{Q_1 \cdot Q_2}}. \quad (1)$$

Normalised coupling factor $\kappa = k/k_{\text{crit}} = k \cdot Q$ (symmetric case):

- **Undercritical** ($\kappa < 1$): single peak at ω_0 ; incomplete energy transfer.
- **Critical** ($\kappa = 1$): maximally flat pass-band (Butterworth); optimal power transfer.
- **Overcritical** ($\kappa > 1$): two peaks at $\omega_{\pm} \approx \omega_0 \left(1 \pm \frac{1}{2} \sqrt{\kappa^2 - 1}\right)$.

Two structural statements [E]

(2.1) Coupling shifts frequencies before splitting them (frequency pulling); the “bare” eigenfrequency is in principle unobservable.

(2.2) The threshold is a ratio: whether a system splits depends on k relative to the loss rates, not on k alone.

Numerically verified by `pruef_328_2_bandfilter.py`: peak formula to $< 10^{-5}$ relative, Butterworth curvature exactly zero at $\kappa = 1$.

.3 Recurrence in Quantum Physics [E]

The correspondence is mathematically exact, not merely analogical:

Communications eng.	Quantum physics	Condition
Overcrit. peak splitting	Rabi splitting, avoided crossing	$g > (\kappa_c + \gamma)/2$
Critical coupling	Critical coupling (resonator QED)	coupling = loss rate
Frequency pulling	Mass/energy renormalisation, Lamb shift	coupling to continuum
Envelope interference	ρ - ω mixing	two poles, shared channel

.4 Measurement as Coupling — the κ -Continuum [B]

Every measurement couples the readout system to the object. Complete decoupling $\equiv$ zero information flow.

Symbolic Proof (von Neumann pointer model)

System $a|+\rangle + b|-\rangle$, Gaussian pointer of width σ , shift $\pm d/2$; coupling parameter $\kappa \equiv d/(2\sigma)$. Exact results:

Exact duality — [B] (Scripts 5 and 9A)

$$V(\kappa) = \langle \phi_+ | \phi_- \rangle = e^{-\kappa^2/2} \quad (\text{back-action / residual coherence}), \quad (2)$$

$$D(\kappa) = \sqrt{1 - e^{-\kappa^2}} \quad (\text{Helstrom distinguishability}), \quad (3)$$

$$D^2 + V^2 = 1 \quad \text{for every pure pointer.} \quad (4)$$

Limits: $\kappa \rightarrow 0$: $(V, D) = (1, 0)$; $\kappa \rightarrow \infty$: $(V, D) = (0, 1)$.

Weak regime: $D \approx \kappa$ (linear), $1 - V \approx \kappa^2/2$ (quadratic).

Not closable — [AXIOM]

"Every measurement apparatus reduces to a pointer model" is a physical model assumption, not provable, only falsifiable. Declared as **[AXIOM]** (model framework).

All quantities V, D, κ reside on the state-transition level of the A271 three-level separation (hardware / state transition / description).

.5 Synchronisation and Locking [E]

Adler Equation

$\dot{\phi} = \Delta\omega - \varepsilon \sin \phi$. Lock-in precisely for $|\Delta\omega| \leq \varepsilon$; lock range $\propto$ coupling ε . Verified numerically (pruef_328_3_einrasten.py, deviation $< 0.1\%$).

Kuramoto Model

Below the critical coupling K_c the ensemble remains incoherent ($r = 0$); above K_c the order parameter $r \propto \sqrt{K - K_c}$ grows continuously — a second-order phase transition. This is **not** a three-way split in the sense of Sec. 2: the Kuramoto transition separates two structural regimes (disorder / growing order), not three. The closer parallel to the band-filter three-way classification is the Adler model (Sec. 5.1): there the lock-range threshold is exactly the coupling amplitude, sharing the same threshold structure as the band-filter critical coupling.

Noise Minimum

The locked state exports entropy and is the least noisy available state — the foundation of every frequency synthesis (PLL).

Thermodynamic Clarification

Phase locking occurs exclusively in open, driven, dissipative systems. “Natural systems tend toward an optimum, not toward uniform distribution” holds for nonequilibrium steady states; the second law remains intact.

.6 Arnold Tongues, KAM, and the Golden Ratio

Qualitative stability [E]

Tongue widths $\Delta\Omega(p/q) \propto K^q$ (Script 9B: slope 1.98 for $q = 2$, 2.69 for $q = 3$). The most irrationally approximable ratio is $\varphi = [1; 1, 1, \dots]$ (KAM). Connects to Doc. 172/188/275 of the corpus.

Quantitative FFGFT connection [K]

From Doc. 008: $m(n) = m_0 e^{\xi n}$, coupling term $g \sim \xi \omega$, hence $g/\omega = \xi$ (n -independent). Effective K in the circle map:

$$K_{\text{eff}} = 2\pi\xi \approx 8.4 \times 10^{-4}. \quad (5)$$

The ξ^2 suppression (Doc. 007: $\sigma_v \sim \xi_0^2 G_F$) corresponds to the K^2 term in the tongue width. At $K_{\text{eff}} \ll 1$: $\Delta\Omega(1/2) \approx 5.6 \times 10^{-8}$. φ lies entirely outside every tongue. **[K]** (Orbifold derivation outstanding.)

.7 Mixing Angles as Coupling Grades — Largely Closed

Dictionary two-circuit $\leftrightarrow$ 2×2 mixing [B]

Hermitian case (Script 1):

$$\lambda_{\pm} = \frac{H_{11}+H_{22}}{2} \pm \sqrt{\left(\frac{H_{11}-H_{22}}{2}\right)^2 + H_{12}^2}, \quad \tan 2\theta = \frac{2H_{12}}{H_{11}-H_{22}}. \quad (6)$$

Non-Hermitian extension (Script 8, exceptional point):

$$g_{\text{EP}} = \frac{|\gamma_1 - \gamma_2|}{4}. \quad (7)$$

For $g < g_{\text{EP}}$: only widths split (undercritical); for $g > g_{\text{EP}}$: frequencies split (Rabi, overcritical).

Band-filter k_{crit} and cavity-QED “strong coupling” are special cases of this EP condition. [B]

Angle	θ [°]	$\sin^2\theta$	$\kappa_{\text{mix}} = \tan 2\theta$
θ_W (Doc. 323: $\sin^2\theta_W = 0.2308$)	28.71	0.2308	1.57
Cabibbo θ_C ($ V_{us} = 0.22452$, Doc. 041)	12.98	0.050	0.49
CKM θ_{13} ($ V_{ub} = 0.00365$)	0.21	≈ 0	0.007
PMNS θ_{12} (Doc. 041: 33.44°)	33.44	0.304	2.34
PMNS θ_{23} (Doc. 041: 49.2°)	49.2	0.573	−6.77
PMNS θ_{13} (Doc. 041: 8.57°)	8.57	0.022	0.31

ξ -Level: θ_W sector closed [K]

$\sin^2 \theta_W = 0.2308$ derived from ξ in Doc. 323 [K]. Hence $\kappa_{\text{mix}}(\theta_W) = 1.5652$ is ξ -derived by composition. Screening against 11 candidate constants: no hit (closest: $e/\sqrt{3}$, 2.7×10^{-3}).

GST texture and H_{12} [B]

Doc.-041 formulas of type $\theta = \arcsin \sqrt{m_1/m_2}$ match the Gatto–Sartori–Tonin texture at leading order ($m_1 \ll m_2$); exactly, the texture gives $\tan \theta = \sqrt{m_1/m_2}$, i.e. $\sin^2 \theta = m_1/(m_1 + m_2)$ (Script 7). The two forms differ by $y^{3/2}/2$ with $y = m_1/m_2$; for $m_1/m_2 = 1/20$, $\arcsin \sqrt{1/20} = 12.92^\circ$ against $\arctan \sqrt{1/20} = 12.60^\circ$ (2.5 %).

$$M = \begin{pmatrix} 0 & \sqrt{m_1 m_2} \\ \sqrt{m_1 m_2} & m_2 - m_1 \end{pmatrix}, \quad \text{eigenvalues } \{-m_1, m_2\}, \quad \tan \theta = \sqrt{m_1/m_2}. \quad (8)$$

$H_{12} = \sqrt{m_1 m_2}$ (geometric mean of masses). [B]

Four check-points from Doc. 041**Check-points for Doc. 190 — author decision required**

- (i) $\delta_{\text{CKM}} = \arcsin(2\sqrt{2}\sqrt{\xi}/3)$ gives 0.011 rad vs. 1.20 rad; candidate: $\arcsin(2\sqrt{2}/3) = 1.231$ rad.
- (ii) $\theta_{13} = \arcsin(\xi^{1/3})$ gives 2.93° vs. 8.57° ; candidate: $\arcsin((25\xi)^{1/3}) = 8.59^\circ$.
- (iii) $|V_{us}|$ with Doc. 006 masses: 5.4 % discrepancy.
- (iv) Doc. 007 $\leftrightarrow$ Doc. 041 tension already declared as `\begin{warning}` and `\begin{speculation}` in the source.

.8 Delimitation**Delimitation from resonance-as-vocabulary**

Schepis (IPI Letters 3(2) 2025; ICIP-2026) uses resonance as connecting vocabulary without computable quantities. This document takes the opposite direction: every bridge is verified by a script and carries a definite status marker. Observers appear only as read-out sub-circuits; no consciousness vocabulary.

.9 Closability Balance

Bridge	Status
R-new-b	[B] Hermitian + non-Hermitian (EP), complete
R-new-c	[K] for θ_W (composition); [B] GST texture; four check-points open
R-new-d	[B] Duality $D^2 + V^2 = 1$, all pure pointers
R-new-a (qual.)	[B] K^q hierarchy, φ quantitatively outside tongues
R-new-a (quant.)	[K] $K_{\text{eff}} = 2\pi\xi$ from Doc. 007/008; orbifold derivation outstanding
[AXIOM]	Reducibility measurement $\rightarrow$ pointer; in principle unprovable

.10 Verification Scripts

Script	Content	Result
pruef_328_1	Dictionary, sympy	[B]
pruef_328_2	Band-filter regimes, numpy	[E] confirmed
pruef_328_3	Adler, Arnold tongues	[E] confirmed
pruef_328_4	κ_{mix} table	Finding
pruef_328_5	Measurement κ , $D^2 + V^2$	[B]
pruef_328_6	ξ -composition, screening	[K]
pruef_328_7	GST texture, corpus check	[B] , check-points
pruef_328_8	EP, non-Hermitian	[B]
pruef_328_9	Pure pointers, K^q	[B]
pruef_328_10	$K_{\text{eff}} = 2\pi\xi$	[K]